

Complete Python Mastery Syllabus (Beginner to Advanced)

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Overview

This syllabus is designed to take a learner from absolute beginner level to advanced professional Python developer level. It includes:

- Core Python Programming
 - Object-Oriented Programming
 - Data Structures & Algorithms
 - File Handling & Automation
 - Web Development
 - API Development
 - Database Integration
 - Data Science & AI
 - DevOps & Cloud Automation
 - Cybersecurity Scripting
 - Testing & Deployment
 - Real-world Projects
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Phase 1 — Python Fundamentals

Module 1: Introduction to Python

Topics

- What is Python?
- History and Features of Python
- Python Applications
- Installing Python
- Setting up VS Code / PyCharm
- Python Interpreter
- Running Python Scripts
- Python Keywords
- Comments and Documentation
- PEP 8 Coding Standards

Hands-on

- Hello World Program
- Calculator Program
- User Input Programs

Mini Project

- Student Information System
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Module 2: Variables and Data Types

Topics

- Variables
- Naming Conventions
- Data Types
- int
- float
- complex
- str
- bool
- NoneType
- Type Conversion
- Dynamic Typing
- Memory Management Basics

Hands-on

- Temperature Converter
 - Currency Converter
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Module 3: Operators

Topics

- Arithmetic Operators
- Comparison Operators
- Assignment Operators
- Logical Operators
- Identity Operators
- Membership Operators
- Bitwise Operators
- Operator Precedence

Practice Problems

- Even/Odd Checker
 - Simple Interest Calculator
 - BMI Calculator
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Module 4: Conditional Statements

Topics

- if Statement
- if-else
- if-elif-else
- Nested Conditions
- Match Case (Python 3.10+)

Hands-on

- Grade Calculator
 - ATM Simulation
 - Login Validation
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Module 5: Loops

Topics

- for Loop
- while Loop
- Nested Loops
- break
- continue
- pass
- Loop Else
- Range Function

Hands-on

- Multiplication Tables
 - Pattern Printing
 - Prime Number Generator
 - Fibonacci Series
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Phase 2 — Data Structures in Python

Module 6: Strings

Topics

- String Creation
- String Indexing
- Slicing
- String Methods
- String Formatting
- f-strings
- Escape Characters
- Regular Expressions Basics

Hands-on

- Password Validator
 - Email Validator
 - Palindrome Checker
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Module 7: Lists

Topics

- List Creation
- Indexing and Slicing
- List Methods
- Nested Lists
- List Comprehension
- Shallow vs Deep Copy

Hands-on

- To-Do Application
 - Shopping Cart
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Module 8: Tuples

Topics

- Tuple Operations
- Packing and Unpacking
- Immutable Nature

- Tuple Methods

Hands-on

- Coordinate System Project
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Module 9: Sets

Topics

- Set Creation
- Set Methods
- Union
- Intersection
- Difference
- Symmetric Difference
- Frozen Sets

Hands-on

- Student Attendance Tracker
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Module 10: Dictionaries

Topics

- Key-Value Pairs
- Dictionary Methods
- Nested Dictionaries
- Dictionary Comprehension
- Iterating Dictionaries

Hands-on

- Contact Book
 - Inventory Management
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Phase 3 — Functions and Modular Programming

Module 11: Functions

Topics

- Defining Functions
- Function Arguments
- Return Values
- Lambda Functions
- Recursive Functions
- Scope of Variables
- Global vs Local Variables
- Docstrings

Hands-on

- Recursive Factorial
 - Mathematical Utilities
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Module 12: Modules and Packages

Topics

- Import Statements
- Creating Modules
- Python Packages
- pip Package Manager
- Virtual Environments
- requirements.txt

Important Libraries

- math
- random
- os
- sys
- datetime
- time

Hands-on

- Utility Package Creation
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Phase 4 — File Handling & Exception Handling

Module 13: File Handling

Topics

- Reading Files
- Writing Files
- Appending Data
- CSV Files
- JSON Files
- File Paths
- Context Managers

Hands-on

- Employee Record System
 - CSV Data Analyzer
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Module 14: Exception Handling

Topics

- try-except
- finally
- else Block
- Custom Exceptions
- Logging
- Debugging

Hands-on

- Robust Banking Application
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Phase 5 — Object-Oriented Programming (OOP)

Module 15: OOP Concepts

Topics

- Classes and Objects
- Constructors
- Instance Variables

- Methods
- Encapsulation
- Inheritance
- Polymorphism
- Abstraction
- Method Overloading
- Method Overriding
- Static Methods
- Class Methods
- Magic Methods

Hands-on

- Library Management System
 - Banking System
 - E-commerce Cart
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Phase 6 — Advanced Python

Module 16: Advanced Concepts

Topics

- Iterators
- Generators
- Decorators
- Context Managers
- Closures
- Comprehensions
- Functional Programming
- map()
- filter()
- reduce()
- itertools
- collections module

Hands-on

- Data Processing Pipelines
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Module 17: Multithreading & Multiprocessing

Topics

- Threading
- Multiprocessing
- Async Programming
- asyncio
- Concurrency vs Parallelism
- Thread Safety

Hands-on

- Multi-threaded Downloader
 - Async API Fetcher
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Phase 7 — Database Programming

Module 18: SQL with Python

Topics

- SQLite
- MySQL Integration
- PostgreSQL Basics
- CRUD Operations
- SQL Queries
- ORM Concepts
- SQLAlchemy

Hands-on

- Student Database System
 - Employee Management App
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Phase 8 — Web Development with Python

Module 19: Flask Framework

Topics

- Flask Basics
- Routing

- Templates
- Forms
- Sessions
- Authentication
- REST APIs
- Flask with Database

Project

- Blog Application
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Module 20: Django Framework

Topics

- Django Architecture
- Models
- Views
- Templates
- Admin Panel
- Authentication
- ORM
- APIs with Django REST Framework

Project

- E-learning Platform
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Phase 9 — APIs & Automation

Module 21: API Development

Topics

- REST APIs
- FastAPI
- Flask APIs
- JSON Handling
- API Authentication
- JWT
- API Testing
- Swagger Documentation

Hands-on

- Task Manager API
 - Authentication API
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Module 22: Automation with Python

Topics

- Web Scraping
- Selenium
- BeautifulSoup
- Email Automation
- Excel Automation
- PDF Automation
- Scheduling Tasks

Hands-on

- LinkedIn Auto Poster
 - Resume Generator
 - Web Scraper
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Phase 10 — Data Science & Machine Learning

Module 23: NumPy

Topics

- Arrays
- Array Operations
- Broadcasting
- Mathematical Operations
- Linear Algebra

Hands-on

- Matrix Operations
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Module 24: Pandas

Topics

- DataFrames
- Data Cleaning
- Filtering
- Grouping
- Merging
- Missing Values
- Data Transformation

Hands-on

- Sales Data Analysis
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Module 25: Data Visualization

Topics

- Matplotlib
- Seaborn
- Plotly
- Charts
- Dashboards

Hands-on

- Interactive Dashboard
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Module 26: Machine Learning

Topics

- Supervised Learning
- Unsupervised Learning
- Regression
- Classification
- Clustering
- Model Evaluation
- Scikit-learn

Algorithms

- Linear Regression

- Logistic Regression
- Decision Trees
- Random Forest
- SVM
- K-Means

Projects

- House Price Prediction
 - Student Performance Prediction
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Module 27: Deep Learning

Topics

- Neural Networks
- TensorFlow
- Keras
- CNN
- RNN
- LSTM
- Transformers Basics

Projects

- Image Classification
 - Handwritten Digit Recognition
 - Chatbot
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Phase 11 — DevOps & Cloud Automation with Python

Module 28: Python for DevOps

Topics

- Linux Automation
- Shell Scripting Integration
- CI/CD Automation
- Jenkins Integration
- Docker SDK
- Kubernetes Python Client
- Terraform Automation

- Ansible Automation

Projects

- Deployment Automation Tool
 - Kubernetes Monitoring Tool
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Phase 12 — Cybersecurity & Ethical Hacking with Python

Module 29: Security Scripting

Topics

- Network Programming
- Socket Programming
- Port Scanner
- Packet Sniffing
- Password Strength Checker
- Cryptography Basics
- Hashing
- Encryption

Projects

- Network Scanner
 - Log Analyzer
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Phase 13 — Testing & Deployment

Module 30: Testing

Topics

- Unit Testing
- PyTest
- Mocking
- Integration Testing
- Test Automation

Hands-on

- API Testing Suite
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Module 31: Deployment

Topics

- Dockerizing Python Apps
- CI/CD Pipelines
- Cloud Deployment
- AWS Basics
- Render Deployment
- Vercel
- Heroku Alternatives
- Nginx Basics

Hands-on

- Deploy Full Stack Application
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Phase 14 — Data Structures & Algorithms using Python

Module 32: DSA

Topics

- Arrays
- Linked Lists
- Stacks
- Queues
- Trees
- Graphs
- Hash Tables
- Searching Algorithms
- Sorting Algorithms
- Recursion
- Backtracking
- Dynamic Programming

Platforms

- LeetCode
 - HackerRank
 - CodeChef
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Phase 15 — Real-Time Projects

Beginner Projects

- Calculator
- Password Generator
- Quiz Application
- Expense Tracker

Intermediate Projects

- Chat Application
- Weather Application
- Library Management System
- Face Detection System

Advanced Projects

- AI Resume Generator
 - DevOps Dashboard
 - AI Chatbot
 - Stock Prediction System
 - Kubernetes Monitoring Platform
 - Real-time Analytics Dashboard
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Recommended Tools

IDEs

- VS Code
- PyCharm
- Jupyter Notebook

Version Control

- Git

- GitHub

Databases

- MySQL
- PostgreSQL
- MongoDB

Deployment

- Docker
 - Kubernetes
 - AWS
 - Render
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Recommended Learning Path

Beginner Level (1–2 Months)

- Python Basics
- Data Structures
- Functions
- File Handling

Intermediate Level (2–4 Months)

- OOP
- APIs
- Flask/Django
- Database Programming

Advanced Level (4–8 Months)

- Machine Learning
 - DevOps Automation
 - Cloud Deployment
 - Deep Learning
 - System Design
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Interview Preparation Topics

Python Interview Questions

- Difference between list and tuple
- Deep copy vs shallow copy
- *args* and **kwargs*
- GIL in Python
- Decorators
- Generators
- MRO
- Memory Management
- Multithreading vs Multiprocessing

Coding Interview Topics

- DSA Problems
 - Problem Solving
 - Time Complexity
 - Space Complexity
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Certification Preparation

Recommended Certifications

- Python Institute Certifications
 - Google Python Certification
 - AWS Developer Associate
 - TensorFlow Certification
 - Microsoft AI Certifications
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Final Capstone Projects

AI + DevOps Projects

- AI-based Deployment Failure Prediction
- Intelligent CI/CD Monitoring System
- Kubernetes Auto-healing Tool
- AI Log Analyzer
- GenAI-based Resume Builder

Full Stack Python Projects

- ERP System
 - LMS Platform
 - E-commerce Platform
 - Real-time Collaboration Tool
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Outcome After Completion

After completing this syllabus, a learner can become:

- Python Developer
 - Backend Developer
 - Full Stack Developer
 - AI/ML Engineer
 - DevOps Engineer
 - Automation Engineer
 - Data Analyst
 - Data Scientist
 - Cloud Engineer
 - Security Engineer
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Best Resources

Documentation

- Python Official Documentation
- Real Python
- GeeksforGeeks
- W3Schools

Practice Platforms

- LeetCode
- HackerRank
- Codewars

GitHub Practice

- Build projects regularly
- Upload projects to GitHub
- Create portfolio repositories

Suggested Weekly Study Plan

Weekdays

- 2 Hours Theory
- 2 Hours Coding Practice
- 1 Hour Problem Solving

Weekends

- Mini Projects
- Revision
- GitHub Upload
- Interview Practice

Final Advice

Mastering Python requires:

- Daily coding practice
- Building projects consistently
- Understanding problem solving deeply
- Learning debugging techniques
- Reading documentation
- Contributing to open source

The fastest way to become industry-ready is:

1. Learn fundamentals strongly
2. Build real projects
3. Deploy applications
4. Practice DSA regularly
5. Learn cloud + DevOps + AI integration